

4 EXPERT SETTINGS

The settings below are available only in the operation level 4 (maintenance and service level).

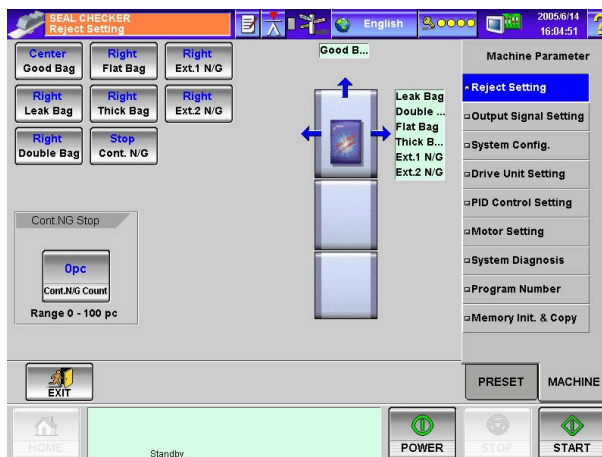
4.1 Machine Parameter screen

The Machine Parameter screen provides access to the screens for performing span adjustment of the seal checker and detail settings of the machines.

This screen is used for selecting a detail setting screen and setting data required for production on that screen.

1. Press the MACHINE TAB on the Preset Parameter screen.

►The Machine Parameter screen appears.



Tabs

- | | |
|----------------------|---|
| Drive Unit Setting: | Changes to the Drive Unit Setting screen. |
| PID Control Setting: | Changes to the PID Control Setting screen. |
| Moter Setting: | Changes to the Motor Setting screen. |
| Memory Initi & Copy: | Changes to the Memory Initialize & Copy screen. |
| System Diagnosis: | Changes to the System Diagnosis screen. |

4.1.1 Drive Unit Setting screen



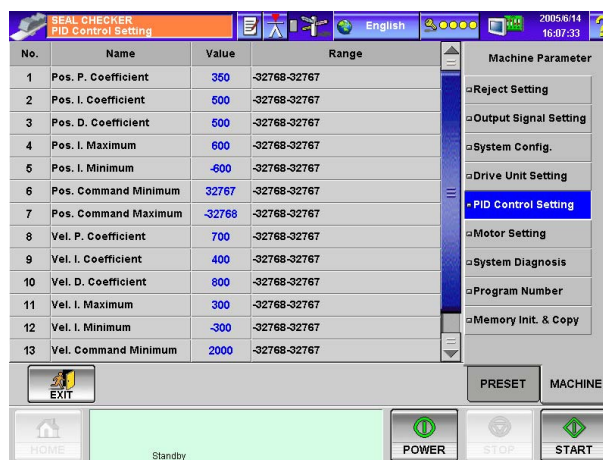
No.	Item	Description
1	Lower Check Conv. Length	Set the lower check conveyor length. To be used for Timing Auto Calculation.
2	Upper Check Conv. Length.	Set the upper check conveyor length. To be used for Timing Auto Calculation.
3	Upper Check Conv. Position.	Set the upper check conveyor position. To be used for Timing Auto Calculation.
4	Photo Sensor Position	Set the photo sensor position. To be used for Timing Auto Calculation.
5	Minimum Bag Length	Set the minimum bag length. The bag length which is shorter than this value cannot be set.
6	Maximum Bag Length	Set the maximum bag length. The bag length which is longer than this value cannot be set.
7	Maximum Standby Height	Set the maximum standby height of upper check conveyor. The standby height which is larger than this value cannot be set.
8	Damper Mode Offset	Set the height offset value which switches from the position mode to the damper mode.
9	Stable Count	Set the stable count. (Not used in the current version.)
10	Stable Height	Set the stable height. (Not used in the current version.)
11	Head Spacer Height	Set the head spacer height. (Not used in the current version.)
12	H Compensation Value	Set the H compensation value. Use this for compensation of the measurement value.
13	Base DA	Set the base DA value. Use this for PID control of the head.
14	JBC-IFC Gap	Set the gap between JBC and IFC. To be used for the pre-reject timing auto calculation.
15	Pre-Reject Distance	Set the distance between the sensor and the air jet of pre-reject part. To be used for the pre-reject timing auto calculation.

No.	Item	Description
16	Pre-Reject Short Coefficient	Set the pre-reject short coefficient. To be used for the pre-reject timing auto calculation.
17	Pre-Reject Long Coefficient	Set the pre-reject long coefficient. To be used for the pre-reject timing auto calculation.
18	Pre-Reject Answer Request	With this setting ON, the pre-reject answer is sent to RCU from the main unit. The status can be checked on the pre-reject result log screen of the hidden screen.
19	Infeed Conv. Length	Set the infeed conveyor length.
20	Reject Distance	Set the distance between the end point of the check conveyor and the spraying point of the rejector air jet

4.1.2 PID Control Setting Screen



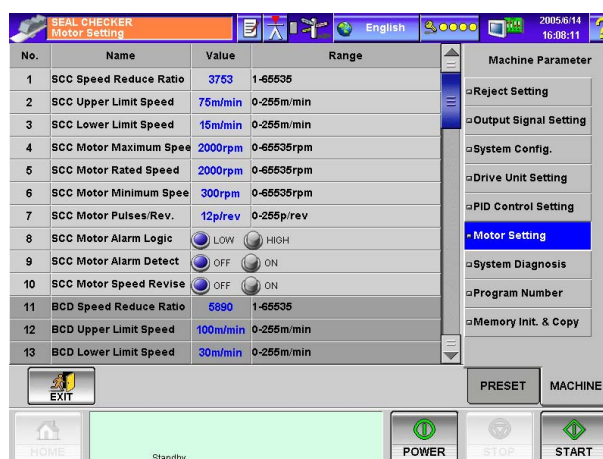
- Do not change any settings stated below. Otherwise oscillation or runaway of the check head may cause.



No.	Item	Description
1	Pos. P. Coefficient	Set the position loop P coefficient (proportional gain).
2	Pos. I. Coefficient	Set the position loop I coefficient (integral gain).
3	Pos. D. Coefficient	Set the position loop D coefficient (derivative gain).
4	Pos. I. Maximum	Set the maximum value of the position loop I term (integral term).
5	Pos. I. Minimum	Set the minimum value of the position loop I term (integral term).
6	Pos. Command Maximum	Set the maximum value of the position loop command output (velocity command).
7	Pos. Command Minimum	Set the minimum value of the position loop command output (velocity command).

No.	Item	Description
8	Vel. P. Coefficient	Set the velocity loop P coefficient (proportional gain).
9	Vel. I. Coefficient	Set the velocity loop I coefficient (integral gain).
10	Vel. D. Coefficient	Set the velocity loop D coefficient (derivative gain).
11	Vel. I. Maximum	Set the maximum value of the velocity loop I term (integral term).
12	Vel. I. Minimum	Set the minimum value of the velocity loop I term (integral term).
13	Vel. Command Maximum	Set the maximum value of the velocity loop command output (torque command).
14	Vel. Command Minimum	Set the minimum value of the velocity loop command output (torque command).

4.1.3 Motor Setting Screen



No.	Item	Description
1	xxx Speed Reduce Ratio	Set the conveyor speed reduction ratio. = Roller rotation count per motor rotation x Roller diameter x 100 π
2	xxx Upper Limit Speed	Set the conveyor upper limit speed. The conveyor speed which is higher than the upper limit cannot be set.
3	xxx Lower Limit Speed	Set the conveyor lower limit speed. The conveyor speed which is lower than the lower limit cannot be set.
4	xxx Motor Maximum Speed	Set the rotation speed when the speed command current 5.00 V is energized to the motor driver. 2200 rpm for HBL motor, 3200 rpm for AXH motor
5	xxx Motor Rated Speed	Set the rated rotation speed as described in the motor data sheet. 2000 rpm for HBL motor, 2500 rpm for AXH motor.
6	xxx Motor Minimum Speed	Set the minimum rotation speed as described in the motor data sheet. 300 rpm for HBL motor, 100 rpm for AXH motor.
7	xxx Motor Pulses/Rev.	Set the single rotation pulse count as described in the motor data sheet. 12 pulse/rev for HBL motor, 30 pulse/rev for AXH motor.
8	xxx Motor Alarm Logic	Set the alarm logic as described in the motor data sheet. LO for HBL motor, HI for AXH motor.

No.	Item	Description
9	xxx Motor Alarm Detect	With this setting ON, the control program detects the motor alarm signal. No detection is made with the setting OFF.
10	xxx Motor Speed Revise	With this setting OFF, the control program revises the motor rotation speed. No revision is made with the setting OFF.

Motor rotation speed [rpm] = (100000 x Conveyor speed [m/min]) / Speed reduction ratio

Motor speed command DA value = (255 x Motor rotation speed [rpm]) / Motor maximum speed

Motor target pulse frequency [Hz] = (Motor Pulses/Rev. x Motor rotation speed [rpm]) / 60

4.1.4 System Diagnosis screen

Seal checker incorporates diagnosis function as the standard specification to display and operate all I/O status of the main unit control board, P-5530A.

The System Diagnosis screen is displayed as a matrix form of 4 (W) x 8 (L), showing the name and the status of I/O. Pressing the button displaying the port name allows to operate the output status. Diagnosis screens are totally 6 pages.

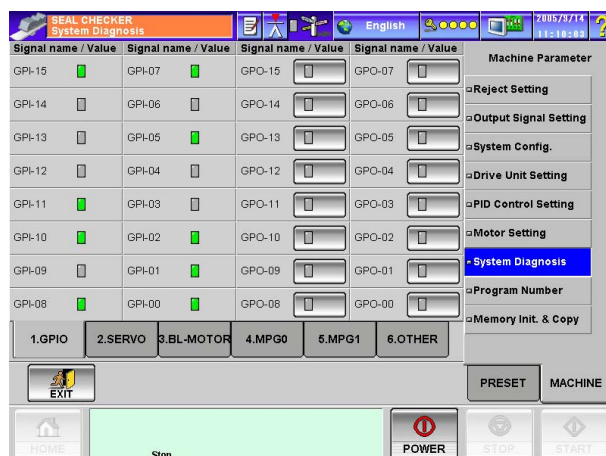
To access the System Diagnosis screen, change the operation level to the level 4 (Maintenance, Service), press the [MACH. SET] key on the MAIN MENU screen and then press the "System Diagnosis" tab.

Page No.	Name	Description
1	GPI0 (GENERAL I/O DIAGNOSIS)	Performs diagnosis of the states of multipurpose input (GPI-15 to GPI-00) and multipurpose output (GPO-15 to GPO-00).
2	SERVO (SERVO I/O DIAGNOSIS)	Performs diagnosis of the servo amplifier I/O state. For diagnosis, the knowledge of servo amplifier I/O is required.
3	BL-MOTOR (BL-MOTOR DIAGNOSIS)	Performs diagnosis of the I/O state of DC brushless motor. The motor can be operated individually for diagnosis.
4	MPG0 (MGP2010(0) DIAGNOSIS)	Performs diagnosis of the state of pulse motor control LSI MPG2010 (Channel 0).Not available in seal checker.
5	MPG1 (MGP2010(1) DIAGNOSIS)	Performs diagnosis of the state of pulse motor control LSI MPG2010 (Channel 1).Not available in seal checker.
6	OTHER (OTHER I/O DIAGNOSIS)	Performs diagnosis of the state of photo sensor and emergency stop input (EMG-03 to EMG-00).

NOTE

- Details of the System Diagnosis screen 2 (SERVO), System Diagnosis screen 4 (MPG0) and System Diagnosis screen 5 (MPG1) are omitted.

4.1.4.1 System Diagnosis screen 1 (GPIO)



I/O name	I/O	Connector	Function/Usage
GPI-15	I	XJ739-3	Interface Terminal block B9 (Packer operating signal)
GPI-14	I	XJ739-4	Interface Terminal block B8 (External error signal 2)
GPI-13	I	XJ739-9	Interface Terminal block B7 (External error signal 1)
GPI-12	I	XJ739-10	Interface Terminal block B6 (Metal detection input signal)
GPI-11	I	XJ738-3	Interface Terminal block B4 (AC horn interlock. Side B ready signal)
GPI-10	I	XJ738-4	Interface Terminal block B3 (AC horn interlock. Side A ready signal)
GPI-09	I	XJ738-9	Interface Terminal block B2 (Operation prohibition signal)
GPI-08	I	XJ738-10	Interface Terminal block B1 (Operation allowance signal)
GPI-07	I	XJ737-2	Fan operation check signal input. Open at the fan stop
GPI-06	I	XJ736-2	Not used
GPI-05	I	XJ735-2	Drop down belt. Auto SW signal input
GPI-04	I	XJ734-2	Drop down belt. Auto SW signal input
GPI-03	I	XJ733-2	ATA busy signal
GPI-02	I	XJ732-2	Enabling signal for packer (upper stream machine) production
GPI-01	I	XJ731-2	Downstream machine operating signal
GPI-00	I	XJ730-2	Air pressure decreasing signal. Open at air pressure decrease
GPO-15	O	XJ720-2	Opening and closing of the error reject solenoid valve 2 (left direction)
GPO-14	O	XJ720-4	Opening and closing of the error reject solenoid valve 1 (left direction)
GPO-13	O	XJ720-6	Opening and closing of the error reject solenoid valve 2 (right direction)
GPO-12	O	XJ720-8	Opening and closing of the error reject solenoid valve 1 (right direction)
GPO-11	O	XJ719-2	Test result output signal 4 (SIG 4)
GPO-10	O	XJ719-4	Test result output signal 3 (SIG 3)
GPO-09	O	XJ719-6	Test result output signal 2 (SIG 2)
GPO-08	O	XJ719-8	Test result output signal 1 (SIG 1)

I/O name	I/O	Connector	Function/Usage
GPO-07	O	XJ728-2	Not used
GPO-06	O	XJ727-2	Drop down belt top/bottom SV
GPO-05	O	XJ726-2	AC horn interlock. Ready OFF alarm at a necessary side
GPO-04	O	XJ725-2	AC horn interlock. Side selection
GPO-03	O	XJ724-2	Opening and closing of the pre-reject solenoid valve
GPO-02	O	XJ723-2	Operating signal (Opening and closing of G2R relay lower right of the control unit)
GPO-01	O	XJ722-2	Opening and closing of the operating magnetic contactor
GPO-00	O	XJ721-2	Opening and closing of the magnetic contactor for servo motor

Application1 of System Diagnosis screen 1 (multipurpose I/O diagnosis)

The following shows the procedure to open and close the pre-reject solenoid valve using the multipurpose I/O diagnosis screen.

1. Press the GPO-03 button to turn ON GPO-03 output signal.
2. Press the GPO-03 button again to turn OFF GPO-03 output signal.

With the procedure shown as above, the pre-reject solenoid valve can be opened and closed.

Application 2 of diagnosis screen 1 (multipurpose I/O diagnosis)

The following shows the procedure to open and close the error reject solenoid valve (right direction) using the multipurpose I/O diagnosis screen.

1. Press the GPO-13 button to turn ON GPO-13 output signal.
2. Press the GPO-13 button again to turn OFF GPO-13 output signal.

With the procedure shown as above, the error reject solenoid valve (right direction) can be opened and closed.

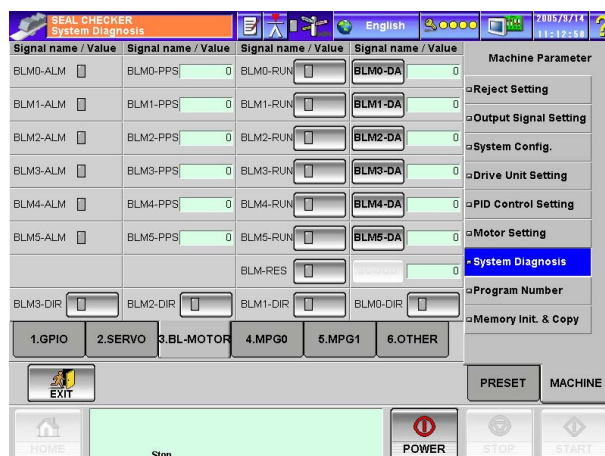
Application 3 of diagnosis screen 1 (multipurpose I/O diagnosis)

The following shows the procedure to open and close the error reject solenoid valve (left direction) using the multipurpose I/O diagnosis screen.

1. Press the GPO-15 button to turn ON GPO-15 output signal.
2. Press the GPO-15 button again to turn OFF GPO-15 output signal.

With the procedure shown as above, the error reject solenoid valve (left direction) can be opened and closed.

4.1.4.2 System Diagnosis screen 3 (BL-MOTOR)



BLMx-ALM is the status of the ALARM signal output from DC brushless motor driver.

BLMx-PPS is the frequency of the SPEED signal output from DC brushless motor driver.

BLMx-RUN is the status of the RUN/BRAKE signal input to DC brushless motor driver.

BLMx-DA is the speed command value input to DC brushless motor driver.

The main unit control board P-5530A is designed to control DC brushless motor with 6 shafts and have the number from 1 to 5.

Seal checker uses DC brushless motor control circuit as in the following table:

Motor No.	Motor Usage
BLM0	Seal check conveyor
BML1	Brush conditioner
BLM2	Infeed conveyor
BLM3	Bag stabilizer (front) driving
BLM4	J-belt conveyor
BLM5	Reject conveyor

The following shows the procedure to operate the reject conveyor only, by using the BL motor I/O diagnosis screen.

1. Turn on GPO-01 (operating magnetic contactor) on the diagnosis screen 1 (multipurpose I/O diagnosis).
2. Turn on BLM5-RUN on the diagnosis screen 3 (BL motor I/O diagnosis).
3. Set the value (0 to 255) of BLM5-DA on the diagnosis screen 3 (BL motor I/O diagnosis).

The procedure shown above allows to operate the reject conveyor only.

Reducing of the value of BLMx-DA decreases the speed and enlarging it increases the speed.

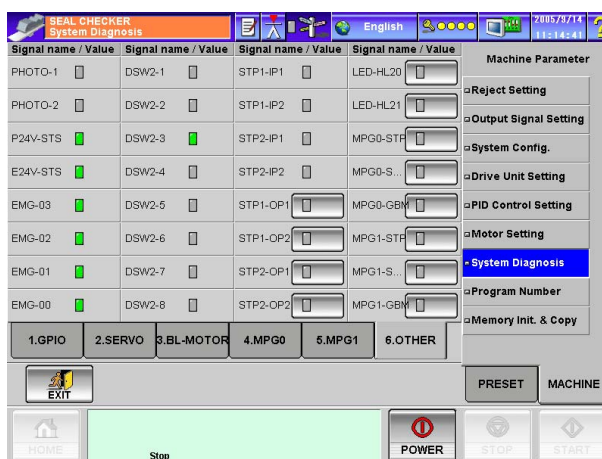
The motor single switch in the control unit is also available. The procedure is as follows:

1. Turn on GPO-01 (operating magnetic contactor) on the diagnosis screen 1 (multipurpose I/O diagnosis)
2. Turn on the motor single switch (upper).
3. Adjust speed with the volume of DC brushless motor driver.

The following table shows the number and the application of motor single switch.

Motor No.	Motor Usage
SS11	Seal check conveyor (upper and lower) driving
SS12	Infeed conveyor driving
SS13	Brush conditioner driving
SS14	Bag stabilizer (front) driving
SS15	J-belt (chute) conveyor driving
SS16	Bag stabilizer (rear) driving
SS21	Reject conveyor driving

4.1.4.3 System Diagnosis screen 6 (OTHER)



I/O name	I/O	Connector	Function/Usage
PHOTO-1	I	XJ744-2	The state of the photo sensor signal on the seal check part (Light income: OFF, Light block: ON)
PHOTO-2	I	XJ745-2	The state of the photo sensor signal on the pre-reject part (Light income: OFF, Light block: ON)
P24V-ST5	I	—	The state of power supply (24 V DC) to the main unit control board
E24V-ST5	I	—	The safety switch operates to turn OFF with power supply (24 V DC) disconnected
EMG-03	I	XJ743-2	Bag stabilizer position (When drawn: OFF, Predefined position: ON)
EMG-02	I	XJ742-2	The state of the door switch on the seal checker part (Door open: OFF, Door close: ON)
EMG-01	I	XJ741-2	Not used

I/O name	I/O	Connector	Function/Usage
EMG-00	I	XJ740-2	The state of emergency stop switch (Normal: ON, at Emergency stop: OFF)

NOTE

- EMG-03 to EMG-00 are the input signals to judge which safety switch is to be pressed. These signals do not work for the safety function (Emergency stop function).

4.1.5 Program Number Screen

The Program Number screen is called up pressing the Program Number index on the Machine Parameter screen.

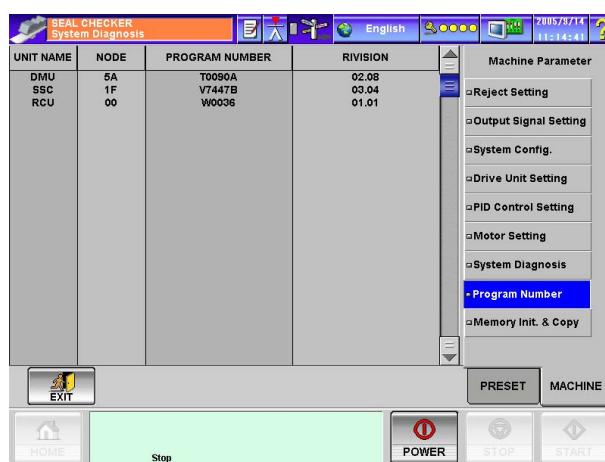


Fig. 4-1 Program Number screen

4.1.6 Memory Inti. & Copy Screen

The memory and memory card are initialized, and the memory data is copied to another memory card.

- The right screen is displayed by pressing the

Memory initialize & copy index. Press the  key, the following confirmation dialog is displayed.

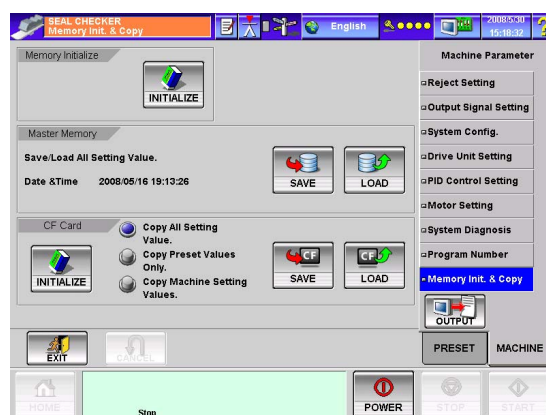
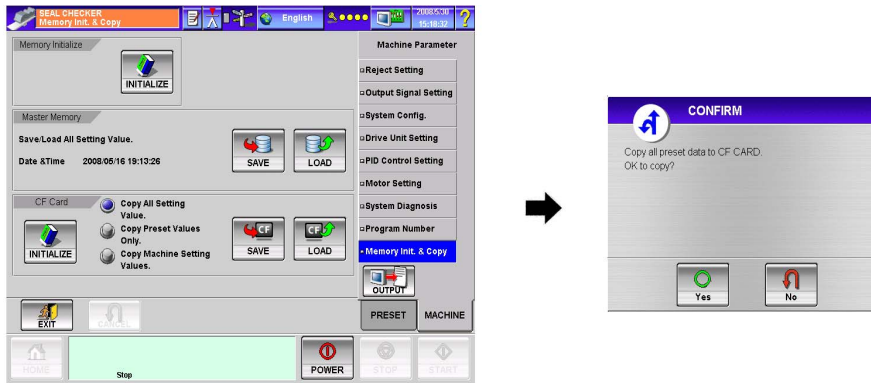


Fig. 4-2 Memory Inti. & Copy screen



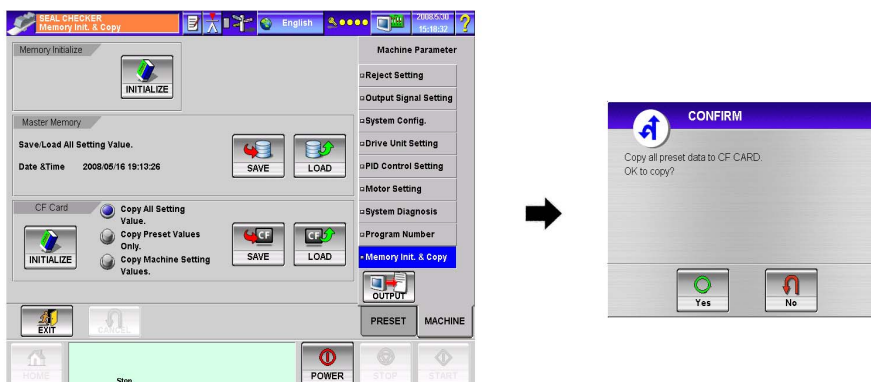
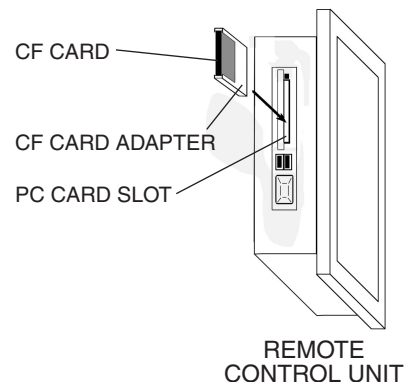
The memory for the equipment is initialized when the [Yes] key is pressed.

TIP

- It is recommendable to copy all the setting data to a CF (compact flash memory card) before initializing the memory.

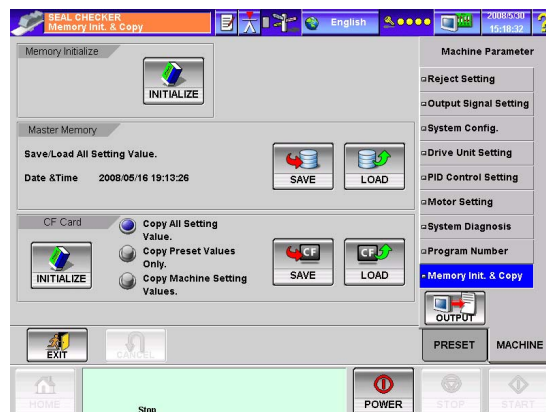
- As in the right figure, insert the compact flash card in the front side of PC card slot in Web-RCU unit.

- Press the  key to initialize the CF. The confirmation message is displayed as shown on the lower right. The CF card is initialized by pressing [Yes] key.



4. Press the [Copy All Setting Value.] button to copy all the setting values from the equipment to the CF card. Although the following screen is displayed, this function does not work as of

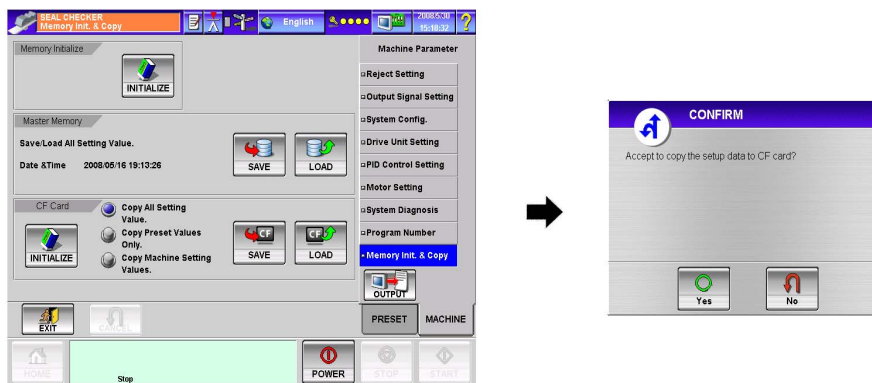
October 2005 even if the  key is pressed.



5. Press the [Copy Preset Values Only.] button to copy only the reservation setting values from the equipment to the CF card. When the  key is touched after the following screen is displayed, the confirmation message is displayed as shown on the lower right. All the reservation setting values are copied from the equipment to the CF card by pressing [Yes] key.

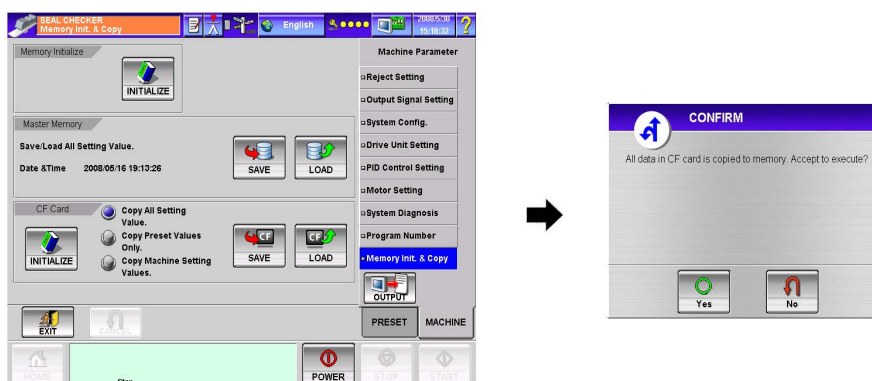


6. Press the [Copy Machine Setting Values.] button to copy only the common setting values from the equipment to the CF card. When the  key is touched after the following screen is displayed, the confirmation message is displayed as shown on the lower right. All the common settings are copied from the equipment to the CF card by touching [Yes] key.



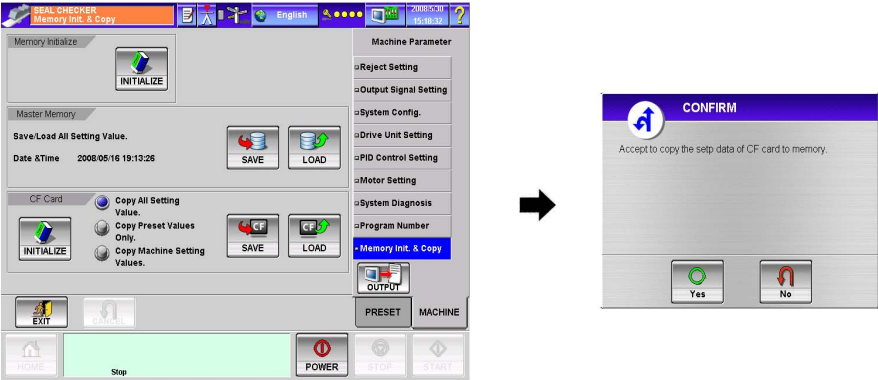
7. Press the [Copy Preset Values Only.] button to copy only the reservation setting values from

the CF card to the equipment. When the  key is touched after the following screen is displayed, the confirmation message is displayed as shown on the lower right. All the reservation setting values are copied from the CF card to the equipment by touching [Yes] key.



8. Press the [Copy Machine Setting Values.] button to copy only the common setting values from the CF card to the equipment. When the

 key is touched after the following screen is displayed, the confirmation message is displayed as shown on the lower right. All the common settings are copied from the CF card to the equipment by touching [Yes] key.



5 APPENDIX

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<MEMO>